Machine Larning Course - Projecte.

Part 3: Association Rule Learning - Discovering Patterns

Market Basket Analysis on Grocery Data

Introduction

The Groceries Dataset consists of supermarket transactions, with each row representing items purchased together. The objective is to discover frequent item combinations in purchases using association rule learning. The Apriori algorithm is applied to identify strong association rules, and the results are visualized using network graphs.

Exploratory Data Analysis

The Groceries Dataset consists of supermarket transactions with 169 unique items, showcasing diverse products. The most frequent items include whole milk, other vegetables, and rolls/buns. The dataset's transaction size distribution is right-skewed, with most transactions containing 4-5 items, indicating typical shopping behavior. The Apriori algorithm identified frequent itemsets with a minimum support threshold of 1%, highlighting strong associations such as {root vegetables, citrus fruit} → {other vegetables} with confidence = 0.58 and lift = 3.02. The Enhanced Association Rules Network Graph effectively visualizes these associations, showing whole milk as a central item linked to dairy and vegetable products, offering strategic insights for retail strategies like cross-promotions and store layout optimizations.

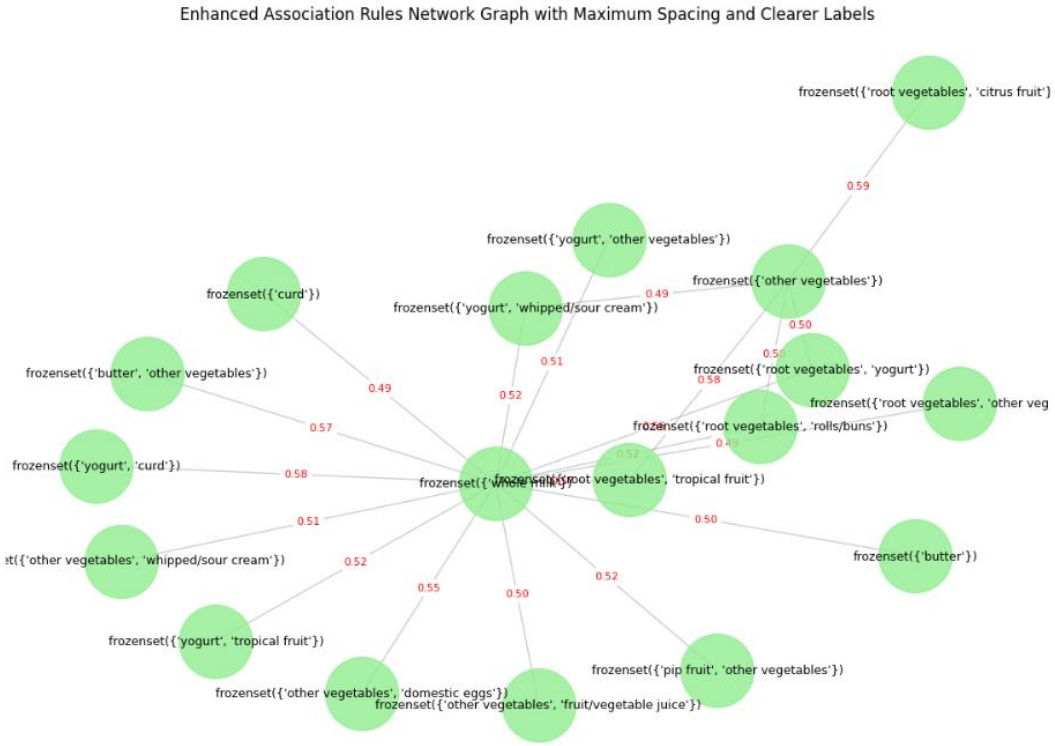
Apriori Algorithm Results

The Apriori algorithm identified frequent itemsets with a 1% support threshold, highlighting strong associations like 'root vegetables' & 'other vegetables' (support = 0.0176) and 'yogurt' & 'curd' (support = 0.0122). The top association rules include {root vegetables, citrus fruit} → {other vegetables} with confidence = 0.58 and lift = 3.02, as well as {yogurt, curd} → {whole milk} with confidence = 0.57 and lift = 2.78, offering valuable insights for retail strategies such as cross-promotions and store layout enhancements.

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
530	(root vegetables, citrus fruit)	(other vegetables)	0.017694	0.193512	0.010372	0.586207	3.029300	1.0	0.006948	1.949012	0.681957	0.051646	0.486920	0.319903
366	(root vegetables, tropical fruit)	(other vegetables)	0.021049	0.193512	0.012304	0.584541	3.020692	1.0	0.008231	1.941197	0.683334	0.060835	0.484854	0.324062
587	(yogurt, curd)	(whole milk)	0.017287	0.255542	0.010067	0.582353	2.278893	1.0	0.005650	1.782505	0.571062	0.038313	0.438992	0.310874
428	(butter, other vegetables)	(whole milk)	0.020033	0.255542	0.011491	0.573604	2.244657	1.0	0.006372	1.745931	0.565833	0.043512	0.427240	0.309285
388	(root vegetables, tropical fruit)	(whole milk)	0.021049	0.255542	0.011999	0.570048	2.230742	1.0	0.006620	1.731492	0.563582	0.045350	0.422463	0.308502

Network Graph Interpretation

The Enhanced Association Rules Network Graph illustrates key item relationships, with whole milk as the central node, showing strong associations with dairy and vegetable items. The confidence scores on edges highlight the strength of associations, with thicker edges representing stronger rules. Notably, high-confidence connections between dairy products and vegetables suggest cross-promotional opportunities for retail strategies.



Conclusion

The Apriori algorithm identified frequent item sets and strong association rules, revealing valuable item correlations through visualization. Key insights support effective marketing and product placement strategies. Future work could explore varying support thresholds and additional metrics to enhance insights.